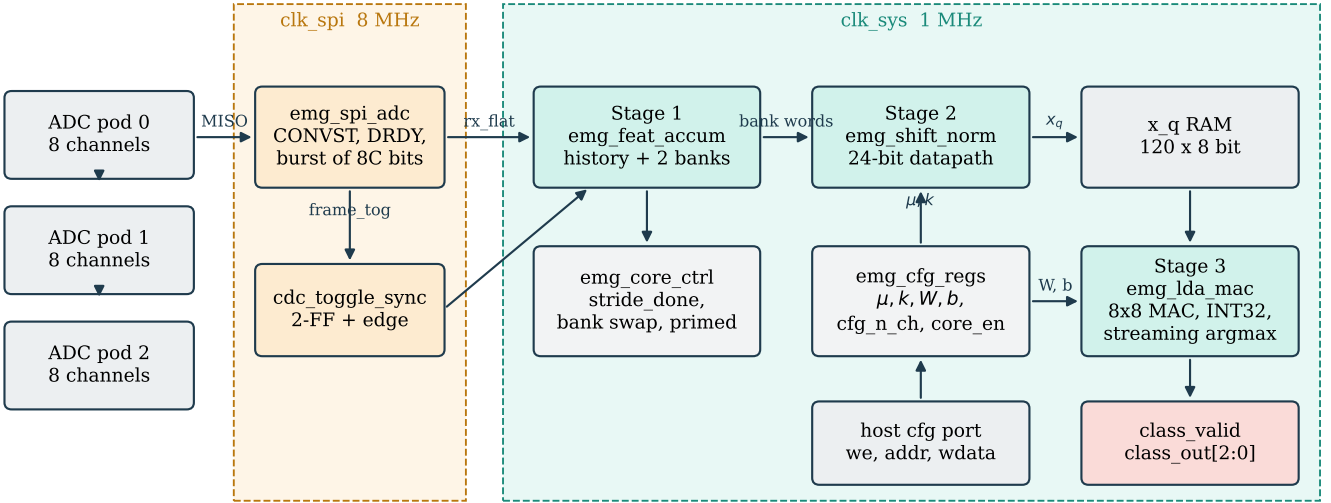
clk_spi 8 MHz
clk_sys 1 MHz
ADC pod 0
8 channels
ADC pod 1
8 channels
ADC pod 2
8 channels
MISO
emg_spi_adc
CONVST, DRDY,
burst of 8C bits
frame_tog
cdc_toggle_sync
2-FF + edge
rx_flat
Stage 1
emg_feat_accum
history + 2 banks
emg_core_ctrl
stride_done,
bank swap, primed
bank words
Stage 2
emg_shift_norm
24-bit datapath
μ, k
emg_cfg_regs
μ, k, W, b,
cfg_n_ch, core_en
host cfg port
we, addr, wdata
x_q
x_q RAM
120 x 8 bit
W, b
Stage 3
emg_lda_mac
8x8 MAC, INT32,
streaming argmax
class_valid
class_out[2:0]